Project Executable Files

Date	29 June 2026
Team ID	xxxxxx
Project Name	xxxxxx
Maximum Marks	3 Marks

Project Executable Files

This section provides all executable and deployable components required to build, test, and run the VBCUA application. The project has been organized with complete source code, documentation, dependency files, testing utilities, and deployment details to ensure that evaluators can easily reproduce and validate the solution.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA / project uses CSV dataset
5	Environment configuration file (.env.example or similar)	NA / No external API
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	YES
10	Demo video or walkthrough (if required)	YES

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not deployed yet (Local development using Streamlit)
Login Credentials (Demo)	Not Required – Application runs locally and does not require authentication.
Platform / Hosting Provider	Localhost (Future deployment: Streamlit Community Cloud / Render / Hugging Face Spaces)
Repository Link	Your team's GitHub repository URL *(replace with your team's repo once available)*
Demo Video Link	To be added after implementation and demonstration are completed.

Step 4: Run Instructions

Clone the Repository

https://github.com/ramyakommuri23/Voice-Based-Analyser/tree/main/Project%20Documentation

Create and Activate a Virtual Environment
python -m venv vbcu_env
Windows

Linux/macOS
source vbcu_env/bin/activate

Install Dependencies
pip install -r requirements.txt

Verify Required Libraries
Run the import verification script to ensure all required libraries are installed correctly.
python test_imports.py
Expected Output:
✅ All libraries imported successfully!
PyTorch Version: ...
Streamlit Version: ...
Librosa Version: ...

Launch the Streamlit Application
streamlit run app.py

Access the Application
Open your browser and visit:
http://localhost:8501
Upload a supported audio file (.wav, .mp3, .m4a, .ogg), select a reference topic, and click **Analyze Concept Understanding** to view the transcript, evaluation metrics, waveform visualization, and downloadable PDF report.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Future Status
1	Audio transcription may take a few seconds for larger audio files due to Whisper model processing.	Future versions will optimize inference speed and support GPU acceleration.
2	The system currently supports a limited set of predefined reference concepts.	Future releases will allow users to add custom concepts and dynamically manage the concept library.
3	The application processes uploaded audio locally and does not maintain long-term user history.	Future versions will integrate a database for user profiles, historical reports, and progress tracking.
4	Speech quality significantly affects transcription and semantic evaluation accuracy.	Future enhancements will include noise reduction, speaker adaptation, and multilingual speech support.
5	The application currently runs as a local Streamlit application.	Planned deployment on Streamlit Community Cloud, Render, or another cloud platform for public access.